

# Azure Technical Deep Dive

Implementing SSL/TLS on High-Availability Infrastructure

## 1. Project Objective

This project demonstrates the implementation of a production-grade, secure web environment. The core focus was transitioning from standard HTTP traffic to fully encrypted HTTPS (Port 443) within a load-balanced architecture, ensuring both security and high availability.

## 2. Technical Architecture

| Component            | Technology Used                           |
|----------------------|-------------------------------------------|
| Compute              | 2x Windows Server 2022 (Availability Set) |
| Traffic Distribution | Azure Standard Load Balancer              |
| Security Management  | Azure Bastion & Network Security Groups   |
| Web Service          | Internet Information Services (IIS)       |

## 3. SSL Implementation Workflow

### Phase A: Guest OS Configuration

- **Certificate Generation:** Created a Self-Signed Certificate via IIS Manager, utilizing the **Web Hosting** store to ensure proper service permissions.
- **Site Binding:** Explicitly bound the Default Web Site to Port 443 using the generated SSL certificate.

- **Guest Firewall:** Configured the Windows Firewall to permit inbound traffic on 443.

```
# PowerShell: Open Port 443 in Windows Firewall
New-NetFirewallRule -DisplayName "Allow-HTTPS" -Direction Inbound -
LocalPort 443 -Protocol TCP -Action Allow
```

## Phase B: Network Security & Load Balancing

- **NSG Rules:** Added a high-priority inbound rule in the Network Security Group to allow TCP Port 443 traffic at the network perimeter.
- **Load Balancing Rule:** Crucially added the "HTTPS Bridge" mapping Port 443 on the Frontend IP to the Backend Pool.
- **Health Probe Optimization:** Transitioned the Health Probe from HTTPS to **TCP Protocol**. This bypassed validation failures caused by the self-signed nature of the certificate while still ensuring port availability.

## 4. Key Troubleshooting Insights

---

**The Missing Link:** Even with valid certificates and open firewalls, traffic failed until the **Load Balancing Rule** was explicitly created. This highlighted that Azure Load Balancers do not automatically pass traffic on 443 without explicit instruction.

**Defense in Depth:** Successfully demonstrated that security must be managed at both the Platform (NSG) and OS (Windows Firewall) layers to achieve connectivity.

## 5. Final Results

---

- Verified 100% encryption success via `https://[Public-IP]`.
- Validated High-Availability failover by deallocating a node and observing continuous service availability.

- Secured administrative access by utilizing Azure Bastion and Serial Console, eliminating public-facing RDP risks.